

July 21, 2026

D. E. Shaw Recruiting Team

Dear D. E. Shaw Recruiting Team,

I am writing to apply for the Quantitative Analyst internship at D. E. Shaw. I am a rising fourth-year Ph.D. student in Computer Science at Carnegie Mellon University, where I study quantum error correction, algorithms, and computational methods for noisy systems.

I am interested in quantitative research because I want my work to have a more direct and measurable impact. While academic research has allowed me to explore fundamental problems, progress in emerging fields often requires substantial resources, which is often not available in academia, and long feedback cycles, which is sometimes frustrating. I am excited by the opportunity to develop rigorous models, test them against real data, use the results to inform practical decisions and see my work have a tangible impact.

My research on quantum error correction has given me solid experience in probability, optimization, statistical inference, algorithm design, and high-performance simulation. I believe these skills are transferable to the role of a quantitative analyst.

I would be excited to bring my mathematical background, research experience, and passion for solving complex problems to D. E. Shaw's collaborative and intellectually rigorous environment. Thank you for your consideration.

Sincerely,

Pengyu Liu